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Prof. Kenneth M. Merz, Jr., Editor-in-Chief

Journal of Chemical Information and Modeling

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Dear Prof. Merz,

We submit our manuscript **“Target breadth and mutation resilience in African-natural-product-inspired antimalarial chemotypes: a computational analysis”** for consideration as an Article in *Journal of Chemical Information and Modeling*.

This work addresses a critical methodological challenge in computational antimalarial drug discovery: evaluating target breadth and mutation resilience as distinct, complementary properties when prioritizing candidates from African natural-product-inspired chemical space, while maintaining clear boundaries between computational evidence and biological activity.

We analyze a 17-member polypharmacology-oriented cohort across four mechanistically diverse antimalarial targets (PfDHFR, PfCRT, PfClpP, and PfATP4) using target-specific structural anchors and a per-target resistance-resilience scoring (RRS) framework. The workflow integrates three evidence layers: chemical-space expansion (65 856 hybrid library molecules $\rightarrow$ 19 913 synthesizable candidates), target-anchored AutoDock Vina docking generating a complete 17×4 scoring matrix (68/68 geometric quality pass rate), and per-target RRS calculated separately for PfDHFR (N51I, C59R, S108N, I164L) and PfCRT (K76T, K76A) mutant panels. The cohort yielded six A*, five B, five C, and one D RRS profiles (range 68.2–111.7), with weak cross-metric associations after multiplicity correction (PNS–RRS $\rho = -0.559$, ACSI–RRS $\rho = -0.132$).

Key Enhancements:

This submission incorporates comprehensive validation, physicochemical characterization, and methodological documentation:

1. **Validation Documentation:** Transparent assessment through three complementary approaches: (a) external benchmark (DEKOIS 2.0, PfDHFR ROC-AUC 0.45, reported honestly as near-chance), (b) enrichment validation (MMV Malaria Box, ROC-AUC 0.924–1.000), and (c) redocking validation (5 ligands, RMSD < 2.0 Å). Complete metrics in Supporting Information Section S12.
2. **Physicochemical Characterization:** All 17 candidates exhibit drug-like properties (100 % Lipinski/Veber compliance, mean QED 0.703, low predicted ADMET risk). Details in Supporting Information Tables S4–S6.

3. **Multi-Parameter Optimization Framework:** Candidate selection utilized a seven-component MPO framework (Vina affinity 35%, DiffDock confidence 25%, QED 20%, ADMET 15%, Lipinski penalty 5%, polypharmacology bonus) with $\text{MPO} \geq 0.40$ threshold. Complete mathematical formulation and weight rationale in Supporting Information Section S10.
4. **Methodological Rigor:** Complete grid box specifications for all four targets, computational efficiency documentation (99.3% cost reduction via centroid-based library reduction), and strict evidence boundaries maintained throughout.

Evidence Boundaries:

The study maintains clear distinctions between computational evidence layers. Results do not claim experimental affinity, confirmed simultaneous target engagement, biological resistance circumvention, or measured $\text{IC}_{50}/\text{EC}_{50}$ values. The workflow produces: (a) four-target docking profiles from which polypharmacology hypotheses can be generated, (b) per-target RRS values summarizing mutation-panel docking ratios, and (c) exploratory cross-metric associations connecting distinct evidence types without conflation.

Accessibility and Impact:

The centroid-based library reduction strategy addresses a critical accessibility gap, particularly for research groups in endemic regions (94% of malaria cases). By reducing computational costs by orders of magnitude while maintaining structural diversity, this workflow enables systematic multi-target screening without high-performance infrastructure, supporting reproducibility and broader participation in computational antimalarial research.

Reproducibility:

Complete computational provenance is provided: source code, candidate manifests, receptor/ligand files, grid configurations, raw docking poses, geometric quality filters, RRS protocols, and machine-readable tables are archived at https://github.com/NanaEngo/Malaria_codesV2. Supporting Information (17 pages, 12 sections, 9 tables, 2 figures) includes complete validation datasets, physicochemical characterization, and methodological specifications.

This integrated approach—connecting chemical-space exploration to target-anchored docking and mutation-aware scoring while maintaining strict evidence boundaries—provides a reproducible template for hypothesis generation in resistance-challenged therapeutic areas. The work will interest the JCIM readership engaged in computational drug discovery, multi-target prioritization strategies, African natural product chemoinformatics, and validation frameworks for docking-derived hypotheses.

This manuscript is original, not under consideration elsewhere, and all authors have reviewed and approved the submission. We declare no competing financial interests.

Sincerely,

Myke Vital Sao Temgoua

(on behalf of all co-authors)